ZCAS UNIVERSITY

SCHOOL OF COMPUTING, TECHNOLOGY AND APPLIED SCIENCES

2025 SESSIONAL EXAMINATIONS

OBJECT ORIENTED PROGRAMMING EXAM (CCS 2121)

INSTRUCTIONS

(1) Time allowed is **Three (3) hours.**

(2) There are Six (6) questions in this paper. **Answer All Questions in Section A and ANY Two Questions from Section B**

(3) Each question carries **Twenty (20)** marks.

(4) Calculators are not allowed.

(5) The total marks for this Exam is **100**

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SECTION A: MANDATORY QUESTIONS

Question one

a) Explain the following terminologies as used in Object Oriented Programming **[2 Marks Each]**

i. Classes
ii. Methods
iii. Objects
iv. Variables
v. Encapsulation
vi. Inheritance
vii. Polymorphism
viii. Dynamic binding
ix. Message Passing

b) Differentiate between Procedural and Objection Oriented Programming. [7 **Mark]**

Question Two

a) Write a class to demonstrate the use of classes, methods, and objects in a practical program. The program should have properties such as name, age, and GPA, as well as a method to print the student's information.

b) Create three Student objects in the Main class and call the printInfo method on each object to print their information. **[20 Marks]**

Question Three

With the use of clear practical examples, provide the syntaxes of the following control structure statements **[5 Marks Each]:**

i. If and else if statement
ii. Switch statement
iii. While Loop
iv. Do while loop
v. For statement

Question Four

You have been employed as a programmer for a shipping company in Lusaka. You are required to create a Java program that calculates the shipping costs for a package based on its weight. The program should take user input for the weight of the package in kilograms.

To calculate the shipping cost, you need to consider different rates based on weight ranges. For example, if the weight is less than or equal to 5 kg, the cost per kilogram is K 2. If the weight

is between 5 kg and 10 kg (inclusive), the cost per kilogram is K 1.5. If the weight is greater than 10 kg, the cost per kilogram is K 1.

a) Use an if-else if statement to determine which weight range the package falls into, and then calculate the total shipping cost accordingly. Once the total shipping cost is calculated, it should be displayed to the user **[10 Marks].**

b) Implement the same program using the switch statement **[10 Marks].**

c) What is the difference between if else statement and switch statement. **[5 Marks]**

Question Five

Consider the following programming question:

Write a program that can be used to compute the distance between two Cartesian Coordinates (For two points $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ in the Cartesian plane). In your program, prompt the user to enter values for x1, x2, y1, and y2. Compute and display the distance using the formula:

$$d = \sqrt{(x2 - x1)^2 + (y2 - y1)^2}$$

Answer the following questions:

(a) Identify the Input, Process and Output required for the program. **[7 Marks]**

(b) Write a full working program matching the stages you have specified in (a). There is only one restriction, your program is not allowed to have local variables. You will lose half the total marks if you ignore this restriction. **[8 Marks]**

Write a Java program that prints all real solutions to the quadratic equation $ax^2 + bx + c = 0$. Read in a, b, c and use the quadratic formula. If the discriminate b^2-4ac is negative, display a message stating that there are no real solutions. **[10 Marks].**

Question Six

You are tasked with developing a program for a library management system. The system should contain information about **books** and **members**. Books have attributes like title, author, ISBN, and genre. Members have attributes like name, ID number, and borrowed books (list of books). **[5 Marks Each]**

Requirements:

Design classes for Book and Member and in each case, implement functionalities for:

i. Adding new books to the library.
ii. Searching for books by title or author.
iii. Registering new members.
iv. Borrowing a book (updating member's borrowed list and book availability).
v. Returning a book (updating member's borrowed list and book availability)

END OF EXAMINATION